

EARTHLAB 2026 — Bayesian Workgroup: Group Tasks

EVERY group must show: ① how much your prior matters (prior-sensitivity) · ② at least one result stated as a probability · ③ one check against data you did not use (posterior-predictive) · ④ your assumptions + how you fit it (conjugate, or MCMC with a convergence check)

GROUP 1 · Sea Surface Temperature · 3 people

`data/bayesian_workgroup/group1_sst/ersstv5_annual_1854_2025.nc` — NOAA ERSSTv5, 2°, 1854–2025

YOUR TASK

The ocean has **mostly** warmed — but not everywhere, not at the same time, and a few regions have even **cooled**. For **every point**, ask: did the trend change in a particular year, and **which way** — and how sure are we? Fit a change-point model in each grid cell and map it.

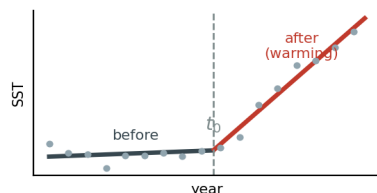
THE MODEL (one line)

$$\mu(t) = \alpha + \beta_1(t - t_0) + \beta_2 \cdot \max(t - t_0, 0)$$

t_0 = the year the trend bends; rate after = $\beta_1 + \beta_2$ (can be + or -)

WHAT TO PRESENT

- > **Map 1 — P(a change point exists):** where are we confident the trend changed?
- > **Map 2 — uncertainty in that year** (t_0 posterior width): your key Bayesian map
- > **Map 3 — rate after the change** (°C / decade; + warming, - cooling)



GROUP 2 · Sea Ice — Change Points · 3 people

`data/bayesian_workgroup/group2_seaice_changepts/sia_series_1901_2024.csv (sia_sep_nh)` — Sept Arctic sea-ice area

YOUR TASK

Arctic September ice has shrunk for a century. Did it decline **steadily**, or did the fall suddenly **accelerate** — and if so, when? Then push that decline forward: when does the Arctic first go ice-free?

THE MODEL

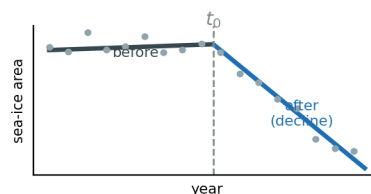
$$\text{one change point: } \mu = \alpha + \beta_1(t - t_0) + \beta_2 \cdot \max(t - t_0, 0)$$

$$\text{two change points: } + \beta_3 \cdot \max(t - t_2, 0)$$

Which does the data prefer? Physical prior: ice declines, so post-slope ≤ 0 .

WHAT TO PRESENT

- > **One change point or two?** (posterior model probabilities / BIC)
- > **The change-point year as a distribution** — does changing the prior move it?
- > **Headline — first ice-free year** (area < 1 Mkm²) as a full distribution: report “**P(ice-free before 2070) = ...**”, not a single year
- > **If time:** Group 1’s three maps, on the gridded file



GROUP 3 · Sea Ice — Forecast · 3 people

`data/bayesian_workgroup/group3_seaice_forecast/sia_series_1901_2024.csv (sia_sep_nh)` — Sept Arctic sea-ice area

YOUR TASK

Begin with what climate models expect — a **prior** on the decline rate (see below). Feed in observations **one year at a time** and watch your belief update. Train on **1979–2019**, then test yourself on **2020–24** (held back): did reality land where you predicted? Finally, forecast 2026 — something next September can prove right or wrong.

THE MODEL + GIVEN PRIOR

$$\mu(t) = \alpha + \beta(t - t_{\text{ref}}) \quad (\beta = \text{decline rate})$$

$\beta \sim \text{Normal}(-0.75, 0.25^2)$ per decade [from IPCC/CMIP]

It’s given — your job is to **justify it** and **test how much it matters**.

WHAT TO PRESENT

- > **The decline-rate posterior tightening** as you add years (sequential updating)
- > **Your 2020–24 prediction vs what really happened** — does the model pass? (watch the recent plateau)
- > **An ice-free-year estimate** from your trend — and whether you trust it after that test
- > **A 2026 forecast** with an uncertainty range

